



Q1. How does a humidity sensor work? Vasilis

A1. Humidity is a measure of the amount of water vapour present in the air. Humidity sensors are low-cost sensitive devices that are used for three primary measurements of humidity: absolute, relative, and specific. Based on these parameters, humidity sensors are classified as capacitive, resistive, and thermal conductivity humidity sensors.

Relative humidity sensors usually contain a humidity sensing element along with a thermistor to measure temperature. Capacitor is used as sensing element in a capacitive humidity sensor whose change in electrical permittivity of the dielectric material is measured to calculate the relative humidity. Low resistivity materials are used in a resistive humidity sensor.

Basic structure of a capacitive humidity sensor is shown in Fig. 1. As you can see, the lower electrodes with two metals (leads) are deposited on the glass substrate and coated with humidity-sensitive polymer material. The capacitor is formed by the upper electrode. Top of the sensing surface is coated with a thin layer of evaporated gold to protect it from ambient contamination or dust and help ensure better condensation.

The high molecular sensitive polymer's dielec-

tric constant changes with relative humidity of the environment. When the humidity changes, capacitance of the humidity sensor changes accordingly. That is, when the relative humidity increases, the humidity-sensitive capacitance increases, and vice versa. The capacitance is usually between 48pF and 56pF. A circuit converts the capacitance to voltage, which corresponds to a change in relative humidity from 0 to 100% RH, and output of the sensor changes linearly from 0 to 1V. Normally, humidity sensor requires a chip or microcontroller to analyse, process, and display the measurement results.

Some humidity sensors available in the market are DHT11, DHT22, AM2302, and SHT71.

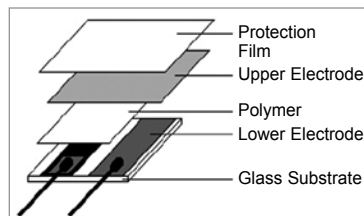


Fig. 1: Structure of capacitive sensor
(Credit: researchgate.net)

Q2. What are the important parameters to check while looking for a low-voltage power supply? Sudhir

A2. Power supply is one the most important equipment for electronics engineers/hobbyists. The choice for power supply depends on the range of power requirement for your applications. The kind of power supplies available include dry and wet batteries, AC to DC, DC to DC, AC to AC, multi-channel, fixed and variable power supplies.

The main generic parameters to check in AC to DC low-voltage power supplies are minimum and maximum voltage and current. Other important parameters include line and load regulation, temperature stability, and short-circuit protection. Note that the voltage requirement for most hobby projects range from 3.3V to 15V DC with current rating from 1mA to 3A.

AC to DC power supply is used for everyday consumer devices such as cell phone chargers, electronics hobbyist and industrial devices. It converts AC power from the grid into DC power and changes the voltage of the power to suit the needs of the devices. Depending on the technologies there are two types: linear and switching power supplies.

Irrespective of the type of power supplies, some common characteristics to look out for in power supplies are:

- Input and output voltage range, value, and accuracy.

- Most conventional power supplies operate with fixed voltage and current ratings. But a variable power supply offers greater flexibility to give a wide range of outputs.

- Required current, and presence or absence of transients
- Load regulation. Typical well-regulated power supplies have load regulations of less than 1%.

- Ripple and noise. Most linear power supplies should have less than 3mV RMS ripple and less than 50mV peak for switching supplies.

- Efficiency, which is the output power divided by the input power, will always be less than 100%.

- Temperature. Modern lab-quality power supplies should have temperature coefficients of under 0.05% per °C. This is usually specified over the operating temperature range, which often is 0 to 40°C.

- Mounting area and height (form factor).
- Whether under-voltage, over-voltage, and over-heating protection required.

- Special environmental/operating conditions. On-board vehicle, space, communications, RF, and other factors.

- Cost of a power supply is also important.